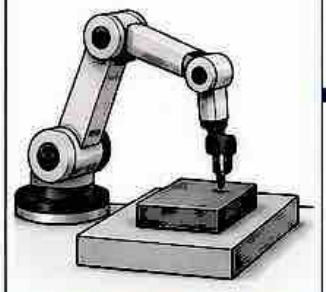
FLOW OF AUTOMATED LASER + MOLTEN SALT PROCESS

A sustainable method for precision material processing and surface treatment

1 MATERIAL LOADING



The raw material is loaded onto the automated platform.

2 MOLTEN SALT APPLICATION



Molten salt is precisely dispensed over the surface to create a protective/reactive layer.

3 LASER IRRADIATION



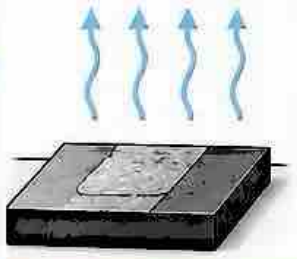
The laser beam interacts with the surface and molten salt, inducing melting, alloying or surface modification.

4 MOLTEN SALT FUNCTION



Molten salt acts as a shield, improves heat transfer and helps in achieving desired surface properties.

5 COOLING & SOLIDIFICATION



The system cools the surface in a controlled manner for proper solidification and microstructure.

6 SALT REMOVAL & CLEANING



Residual molten salt is removed using automated washing or rinsing systems.

AUTOMATION & CONTROL



- Robotic handling
- Precise salt dispensing
- Laser power control
- Real-time monitoring (temperature, feedback)
- Closed-loop system for consistent results

PROCESS BENEFITS

- High precision & repeatability
- Reduced oxidation & defects
- Improved surface quality
- Energy efficient
- Environmentally safer (salt is reusable)



Recyclable Molten Salt

TYPICAL APPLICATIONS

- Surface hardening
- Corrosion resistance
- Cladding & alloying
- Aerospace components
- Medical implants
- Tool & die manufacturing

THINGS EXTRACTED / OBTAINED FROM THIS PROCESS

- Enhanced Surface Layer (hardened / modified layer)
- Alloyed / Cladded Material (high performance coating)
- Refined Microstructure (improved grain structure)
- Low Oxidation & Clean Surface
- Recyclable Molten Salt (recovered and reused)



SUMMARY

In this automated process, molten salt is applied to the material surface and a high-energy laser beam is used to melt and modify the surface. The molten salt protects the surface, enhances heat transfer and helps achieve superior quality. Automation ensures precision, consistency and efficiency.